



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN
GRADE/GRAAD: 12**

**TECHNICAL MATHEMATICS P2/TEGNIJSE WISKUNDE V2
2022
MARKING GUIDELINES/NASIENRIGLYNE**

MARKS/PUNTE: 150

CODE/ KODE	EXPLANATION/VERDUIDELIKING
A	Accuracy/Akkuraatheid
AO	Answer only/Slegs antwoord
CA	Consistent accuracy/Volgehoue akkuraatheid
I	Identity/Identiteit
NPU	No penalty for omitting units/Geen penalisering vir eenhede weggelaat nie
R	Rounding/Afronding
RE	Reason/Rede
S	Simplification/Vereenvoudiging
SF	Substitution in correct formula/Vervanging in korrekte formule
ST/RE	Statement with reason/Bewering met rede
F	Correct formula/Korrekte formule
PR	Penalty for rounding/Penalisering vir afronding

**These marking guidelines consist of 22 pages.
Hierdie nasienriglyne bestaan uit 22 bladsye.**

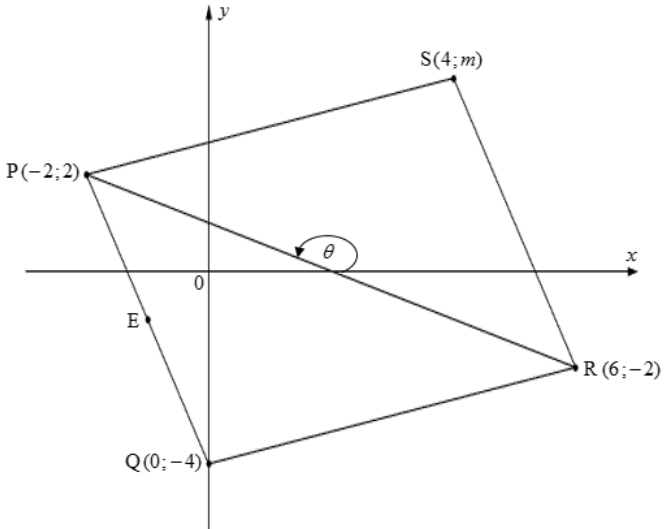
NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- The method of consistent accuracy marking must be applied in all aspects of the marking guidelines where applicable as indicated with the marking code CA
- If a candidate strikes off a response to a question and does not attempt the question again, then the struck off question should be marked

LET WEL:

- Indien 'n kandidaat 'n vraag TWEE keer beantwoord, sien slegs die EERSTE poging na.
- Die metode van volgehoue akkuraatheid-nasien moet, waar moontlik, tot alle aspekte van die nasienriglyne toegepas word soos aangedui deur the nasienkode CA.
- Indien 'n kandidaat 'n antwoord deurhaal en nie poog om die vraag weer te beantwoord dan moet die deurgehaalde antwoord gemerk word.

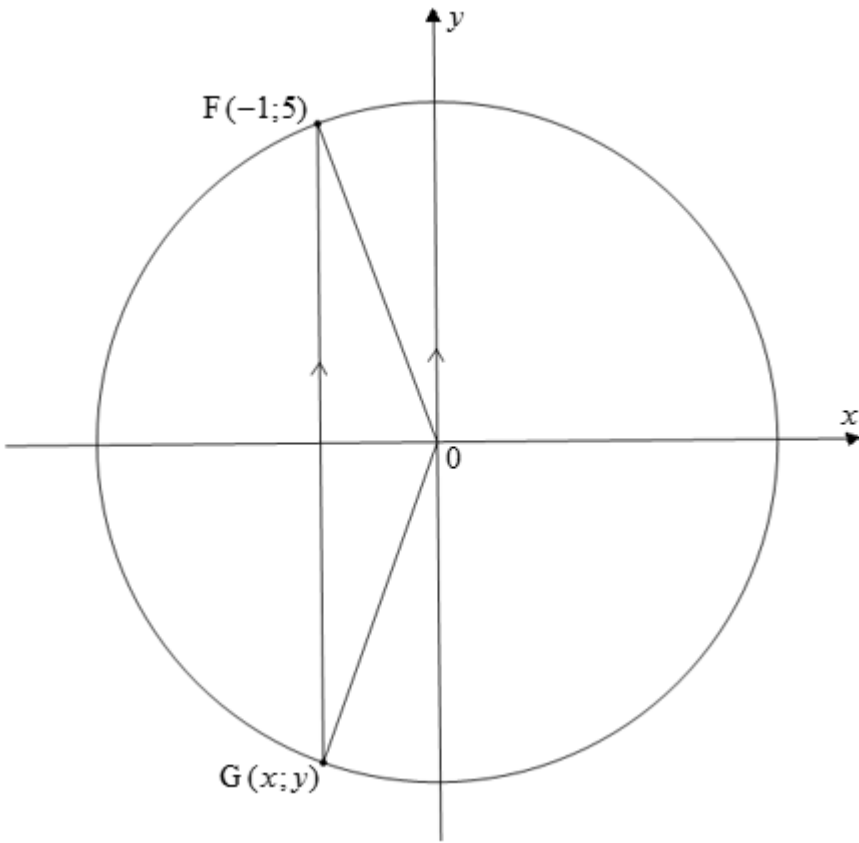
QUESTION/VRAAG 1

		
1.1.1	$m_{PR} = \frac{y_P - y_R}{x_P - x_R}$ $= \frac{2 - (-2)}{-2 - 6}$ $= -\frac{1}{2}$	✓ SF A ✓ gradient/gradiënt CA AO Full marks (2)
1.1.2	$\tan \theta = m = -\frac{1}{2}$ ref. $\angle \approx 26,57^\circ$ $\therefore \theta \approx 180^\circ - 26,57^\circ$ $\approx 153,43^\circ$	CA from/ vanaf Q/V1.1.1 ✓ SF CA ✓ ref angle/verw. hoek CA ✓ value of angle/waarde van hoek CA AO Full marks/Volpunte NPR (3)

1.1.3	$QR = \sqrt{(x_Q - x_R)^2 + (y_Q - y_R)^2}$ $= \sqrt{(0 - 6)^2 + (-4 - (-2))^2}$ $= \sqrt{40} \text{ OR } 2\sqrt{10}$	✓ SF A ✓ Length of QR in surd form/ Lengte in wortelvorm CA AO Full marks/Volpunte
		(2)
1.1.4	$E \left(\frac{x_P + x_Q}{2}; \frac{y_P + y_Q}{2} \right)$ $E \left(\frac{-2 + 0}{2}; \frac{2 + (-4)}{2} \right)$ $E (-1; -1)$ OR/OF $x_E = \frac{x_1 + x_2}{2}, y_E = \frac{y_1 + y_2}{2}$ $x_E = \frac{-2 + 0}{2}, y_E = \frac{2 + (-4)}{2}$ $E (-1; -1)$	✓ x-value/waarde A ✓ y-value /waarde A ✓ x-value/waarde A ✓ y-value /waarde A [Penalty of one mark if not simplified/ Penaliseer met een punt indien nie vereenvoudig nie]
		(2)
1.1.5	$m_{PQ} = \frac{2 - (-4)}{-2 - 0} = -3$ $m_{SR} = -3$ $y = -3x + c$ $-2 = -3(6) + c$ $c = 16$ $\therefore y = -3x + 16$ OR/OF $m_{PQ} = \frac{2 - (-4)}{-2 - 0} = -3$ $y - y_1 = m(x - x_1)$ $y - (-2) = -3(x - 6)$ $y = -3x + 18 - 2$ $\therefore y = -3x + 16$	✓ m_{PQ} A ✓ m_{SR} CA ✓ substitution/vervanging CA ✓ equation/vergelyking CA OR/OF ✓ m_{PQ} A ✓ m_{SR} CA ✓ substitution/vervanging CA ✓ equation/vergelyking CA
		(4)

1.1.6	$y = -3x + 16$ $= -3(4) + 16$ $= 4$ OR/OF $m_{RS} = -3$ $\frac{m + 2}{4 - 6} = -3$ $m + 2 = 6$ $\therefore m = 4$	✓ SF CA ✓ y coordinate/-koördinaat CA OR/OF ✓ subst./verv. (4; m) CA ✓ value of m/waarde van m CA AO Full marks/Volpunte (2)
1.2	$m_{QR} = \frac{-2 - (-4)}{6 - 0}$ $= \frac{1}{3}$ $m_{PQ} \times m_{QR} = -3 \times \frac{1}{3} = -1$ OR/OF $PQ^2 = (-2 - 0)^2 + (2 - (-4))^2 = 40$ $PR^2 = (-2 - 6)^2 + (2 - (-2))^2 = 80$ $PR^2 = PQ^2 + QR^2$	✓ gradient of/gradiënt van QR A ✓ product/produk A OR/OF ✓ length of/lengte van PQ and PR A ✓ Pyth. theorem PR A (2) [17]

QUESTION/VRAAG 2

			
2.1.1	$G(-1;-5)$	$\checkmark x$ $\checkmark y$	A A (2)
2.1.2(a)	$m_{OF} = -5$	$\checkmark$ gradient/ <i>gradiënt</i>	A (1)
2.1.2(b)	$m_{\text{tang}} = \frac{1}{5}$ $y = mx + c$ OR/OF $y - y_1 = m(x - x_1)$ $5 = \frac{1}{5}(-1) + c$ OR/OF $y - 5 = \frac{1}{5}(x - (-1))$ $c = \frac{26}{5}$ $\therefore y = \frac{1}{5}x + \frac{26}{5}$ or/of $y = 0,2x + 5,2$ OR/OF	$\checkmark$ gradient/ <i>gradiënt</i> $\checkmark$ substitution/ <i>vervang</i> $\checkmark$ equation/ <i>vergelyking</i>	CA A CA OR/OF

$$r^2 = (-1)^2 + (5)^2$$

$$\therefore r^2 = 26$$

$$x_1 \cdot x + y_1 \cdot y = r^2$$

$$-x + 5y = 26$$

$$5y = x + 26$$

$$\therefore y = \frac{1}{5}x + \frac{26}{5} \text{ or / of } y = 0,2x + 5,2$$

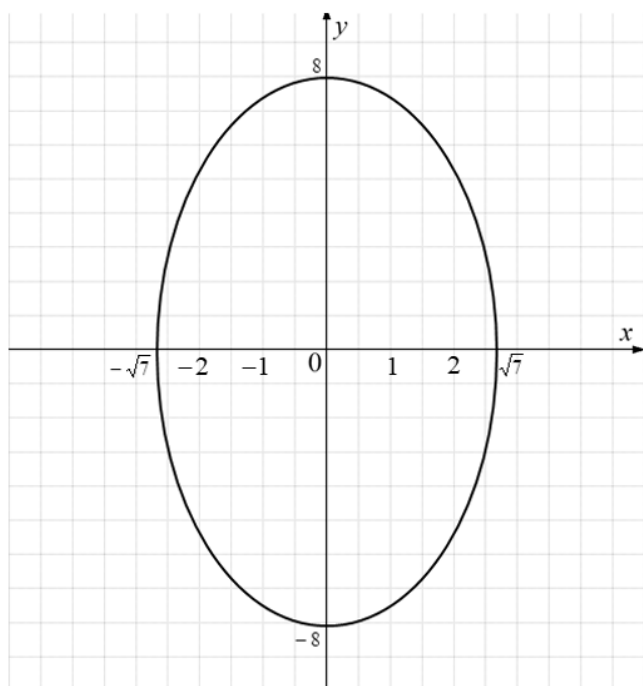
✓ value of r^2 /waarde van r^2 **A**

✓ substitution/vervanging **CA**

✓ equation/vergelyking **CA**

(3)

2.2



✓ both x-intercepts/
beide x-afsnitte **A**

✓ both y-intercepts
beide y-afsnitte **A**

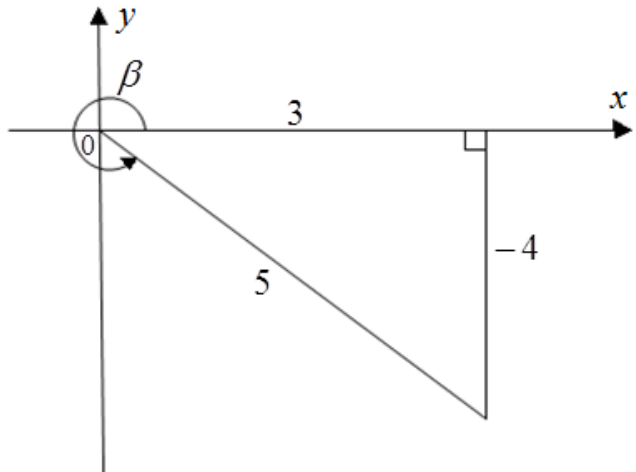
✓ elliptical shape/eliptiese vorm
CA

(3)

[9]

QUESTION/VRAAG 3

3.1.1	$\cot(P - Q)$ $= \cot(71^\circ - 42^\circ)$ $= \frac{1}{\tan(29^\circ)}$ $\approx 1,80$	$\checkmark$ substitution/vervangings A $\checkmark$ I A $\checkmark$ 1,80 A AO Full marks/Volpunte (3)
3.1.2	$\frac{\cos Q}{\sec P}$ $= \frac{\cos 42^\circ}{\sec 71^\circ}$ $= \cos 42^\circ \div \frac{1}{\cos 71^\circ}$ $\approx 0,24$	$\checkmark$ substitution/vervangings A $\checkmark$ I A $\checkmark$ 0,24 CA (3)

3.2	$3\sec\beta - 5 = 0$ $\sec\beta = \frac{5}{3} \quad \text{OR/OF} \quad \cos\beta = \frac{3}{5}$ $(5)^2 = (3)^2 + y^2$ $y^2 = 16$ $y = \pm 4$ $\therefore y = -4$  $\sin^2\beta - \cos^2\beta$ $= \left(-\frac{4}{5}\right)^2 - \left(\frac{3}{5}\right)^2$ $= \frac{7}{25}$	✓ sec/cos ratio/-verhouding A ✓ substitution/vervanging A ✓ correct y-value/korrekte y-waarde CA ✓ diagram CA ✓ sin ratio/-verhouding CA ✓ cos/-verhouding CA ✓ simplification/vereenv. CA (7)
3.3	$\cos 2x - \tan 29^\circ = 0$ $\cos 2x = \tan 29^\circ$ $\cos 2x = 0,554309051$ $\text{ref. } \angle \square 56,34^\circ$ $2x = 56,34^\circ \text{ or } 2x = 360^\circ - 56,34^\circ$ $\therefore x \square 28,17^\circ \text{ or } x \square 151,83^\circ$	✓ $\cos 2x = \tan 29^\circ$ A ✓ ref. angle/verwys. hoek CA ✓ correct quadrants/korrekte kwadrante A ✓ $28,17^\circ$ CA ✓ $151,83^\circ$ CA (5)
	[18]	

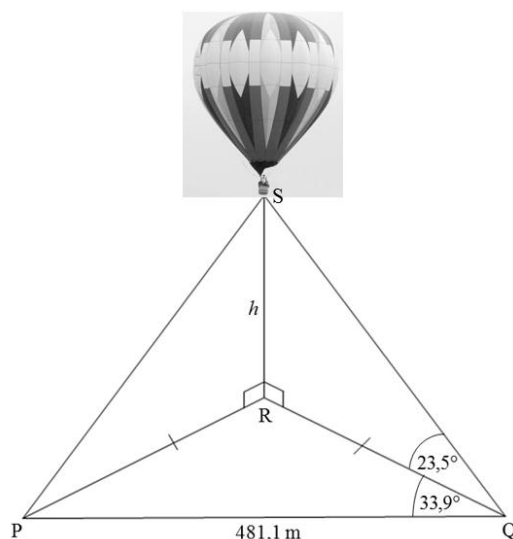
QUESTION/VRAAG 4

4.1	$\cot^2 A \cdot \sin^2 A + \cos^2 A \cdot \tan^2 A$ $= \frac{\cos^2 A}{\sin^2 A} \cdot \sin^2 A + \cos^2 A \cdot \frac{\sin^2 A}{\cos^2 A}$ $= \cos^2 A + \sin^2 A$ $= 1$	✓ I for/ vir $\frac{\cos^2 A}{\sin^2 A}$ A ✓ I for/ vir $\frac{\sin^2 A}{\cos^2 A}$ A ✓ S CA ✓ I for/ vir 1 CA (4)
4.2	$\frac{\sin^2(\pi + \theta) + \cos(180^\circ - \theta) \cdot \sec(360^\circ - \theta)}{\tan(2\pi - \theta) \cdot \cot(180^\circ + \theta)} = \cos^2 \theta$ $\text{LHS} = \frac{\sin^2 \theta + (-\cos \theta) \cdot \sec \theta}{-\tan \theta \cdot \cot \theta}$ $= \frac{\sin^2 \theta - \cos \theta \cdot \frac{1}{\cos \theta}}{-\tan \theta \cdot \frac{1}{\tan \theta}}$ $= \frac{\sin^2 \theta - 1}{-1}$ $= \frac{-(1 - \sin^2 \theta)}{-1} = \frac{-\cos^2 \theta}{-1}$ $= \cos^2 \theta$ ∴ LHS = RHS	✓ $\sin^2 \theta$ A ✓ $-\cos \theta$ A ✓ $\sec \theta$ A ✓ $-\tan \theta$ A ✓ $\cot \theta$ A ✓ I for/ vir $\frac{1}{\cos \theta}$ A ✓ I for/ vir $\frac{1}{\tan \theta}$ A ✓ simplification/vereenv. CA ✓ I for/ vir $-\cos^2 \theta$ A (9)
		[13]

QUESTION/VRAAG 5

5.1	180°	✓ period/periode A (1)
5.2		f : ✓ shape/vorm A ✓ intercepts/afsnitte ✓ turning point/ draaipunt A g : ✓ shape/vorm A ✓ intercepts/afsnitte A ✓ end-points/ eindpunte A (6)
		[7]

QUESTION/VRAAG 6



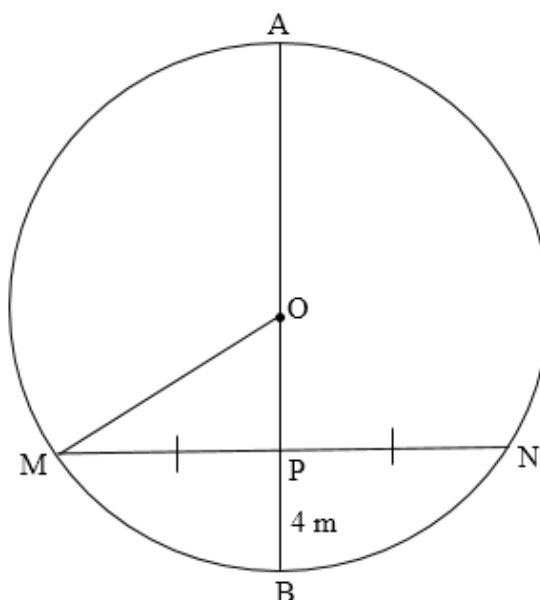
6.1	$\hat{P}RQ + 33,9^\circ + 33,9^\circ = 180^\circ$ $\therefore \hat{P}RQ = 112,2^\circ$	✓ M A ✓ size of/grootte van $\hat{P}RQ$ CA AO Full marks/Volpunte (2)
6.2	$\frac{RQ}{\sin 33,9^\circ} = \frac{481,1}{\sin 112,2^\circ}$ $RQ = \frac{481,1 \cdot \sin 33,9^\circ}{\sin 112,2^\circ}$ $\therefore RQ \approx 289,81 \text{ m}$ OR/OF $RQ^2 \approx PR^2 + (481,1)^2 - 2(481,1)PR \cos 33,9^\circ$ $962,2PR \cos 33,9^\circ \approx 231\,457,21$ $PR \approx \frac{231\,457,21}{962,2 \cos 33,9^\circ}$ $RQ \approx 289,81 \text{ m} \qquad PR = RQ$	✓ SF CA ✓ RQ subject/ -onderwerp A ✓ value of/waarde van RQ CA OR/OF ✓ SF CA ✓ PR subject/ -onderwerp A ✓ value of/waarde van RQ CA (3)
6.3	$\tan 23,5^\circ = \frac{h}{289,81}$ $h = 289,81 \cdot \tan 23,5^\circ$ $\therefore h = 126 \text{ m}$	✓ tan ratio/-verhouding A ✓ h subject/-onderwerp CA ✓ value of/waarde van h CA PR (3)

6.4	$\text{Area of } \triangle QPR = \frac{1}{2}(481,1)(289,81)\sin 33,9^\circ$ $\square 38\,882,53 \text{ m}^2$ OR/OF $\text{Area of } \triangle QPR = \frac{1}{2}(289,81)(289,81)\sin 112^\circ$ $\square 38\,881,86 \text{ m}^2$	✓ SF CA ✓ area value/ <i>oppervlaktewaarde</i> CA from/vannuit 6.1 and 6.2 OR/OF ✓ SF CA ✓ area value/ <i>oppervlaktewaarde</i> CA from/vannuit 6.1 and 6.2 (2) [10]
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QUESTION/VRAAG 7

7.1	Perpendicular to the chord/ <i>Loodreg op die koord</i>	✓ answer/antw. A (1)
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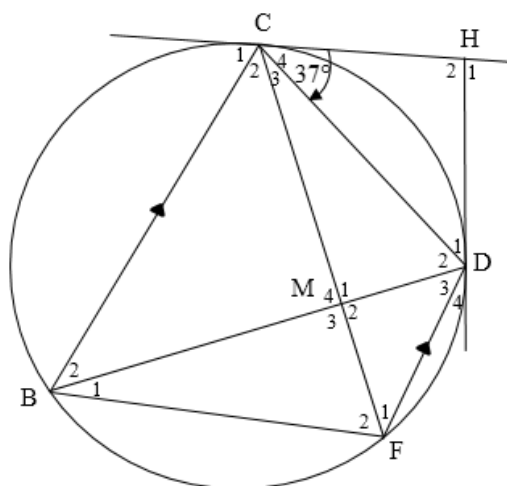
7.2



7.2.1	Diameter// <i>middel lyn</i> : $16\text{m} + 4\text{m} = 20\text{m}$ $OM = 10 \text{ m}$	✓ ST A (1)
7.2.2	$\hat{OPM} = 90^\circ$ (line from centre to midpoint of chord/ <i>lyn van middelpunt na mdpt van koord</i>) $OM^2 = OP^2 + MP^2$ (pythagoras) $10^2 = 6^2 + MP^2$ $MP^2 = 100 - 36$ $MP = 8 \text{ m}$ OR/OF	✓ M A ✓ 6 A ✓ SF CA ✓ S CA ✓ Value of/Waarde van MP CA OR/OF

	$4h^2 - 4hd + x^2 = 0$ $4(4)^2 - 4(4)(20) + MN^2 = 0$ $MN^2 = 256$ $MN = 16$ $MP = 8 \text{ m}$	✓ F A ✓ SF A ✓ simplification/vereenv. A ✓ Value of/Waarde van MN A ✓ Value of/Waarde van MP A (5)
		[7]

QUESTION/VRAAG 8



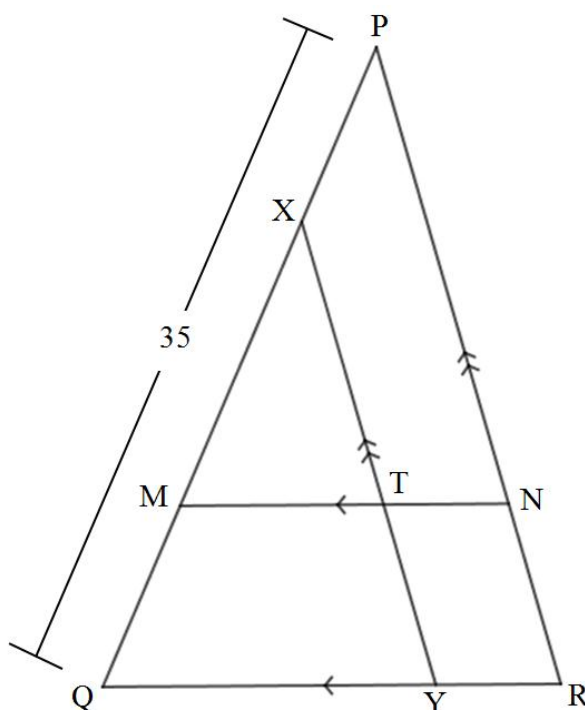
8.1	$\hat{D}_1 = 37^\circ$ (tangents from the same point/ <i>raaklyne van dieselfde punt</i>) $\hat{H}_1 = 74^\circ$ (exterior/ <i>buite</i> $\angle$ in Δ) OR/OF $\hat{C}_4 = \hat{B}_2 = 37^\circ$ (tan-chord th./ <i>rklyn-koord st</i>) $\hat{D}_1 = \hat{B}_2 = 37^\circ$ (tan-chord th./ <i>rklyn-koord st</i>) $\hat{C}_4 = \hat{D}_1 = 37^\circ$ $\hat{H}_1 = 74^\circ$ (exterior/ <i>buite</i> $\angle$ in Δ)	$\checkmark$ ST A $\checkmark$ RE A $\checkmark$ ST A $\checkmark$ RE A OR/OF $\checkmark$ ST A $\checkmark$ RE A $\checkmark$ ST A $\checkmark$ RE A (4)
8.2	$\hat{C}_4 = \hat{F}_1 = 37^\circ$ (tan-chord th./ <i>rklyn-koord st</i>) $\hat{C}_2 = \hat{F}_1 = 37^\circ$ (alt/ <i>verw.</i> $\angle^s$, $BC \parallel FD$)	$\checkmark$ ST A $\checkmark$ RE A $\checkmark$ ST CA $\checkmark$ RE A (4)
8.3	$\hat{C}_2 = \hat{D}_3 = 37^\circ$ ($\angle$ s in the same segment/ <i>in dieselfde segment</i>) $\hat{F}_1 = 37^\circ$ (proven/ <i>bewys</i> in 8.2) $\therefore MD = MF$ (sides opp./ <i>sye teenoor</i> = $\angle$) OR/OF $\hat{F}_1 = \hat{B}_2$ ($\angle$ s in the same segment/ <i>in dieselfde segment</i>) $\hat{D}_3 = \hat{B}_2$ (alt/ <i>verw.</i> $\angle$ s , $BC \parallel FD$) $\hat{F}_1 = \hat{D}_3$ $\therefore MD = MF$	$\checkmark$ ST CA $\checkmark$ RE A $\checkmark$ RE A OR/OF $\checkmark$ ST CA $\checkmark$ RE A $\checkmark$ ST A (3)

8.4	$\hat{M}_1 = 74^\circ$ (exterior/buite $\angle$ of Δ)	✓ ST	CA
	$\hat{H}_1 = \hat{M}_1$ (exterior/buite $\angle$ of Δ)	✓ RE	A
	$\therefore$ CHDM is a cyclic quad. (exterior $\angle$ = op pint $\angle$)	✓ ST	CA
	CHDM is 'n koordevierhoek./ (buite $\angle$ = teenst $\angle$)	✓ RE	A
	OR/OF	OR/OF	
	$\hat{H}_2 = 180^\circ - 74^\circ = 106^\circ$ sum of/som van $\angle$ s of Δ	✓ ST	CA
	$\hat{M}_1 = 74^\circ$ ext./buite $\angle$ of Δ	✓ RE	A
	$\hat{H}_2 + \hat{M}_1 = 180^\circ$	✓ ST	CA
	$\therefore$ CHDM is a cyclic quad. (opp. $\angle$ s are supplementary)	✓ RE	A
	CHDM is 'n koordevierhoek. (teenst $\angle$ e is supplim.)		(4)
			[15]

QUESTION/VRAAG 9

9.1	Divides the other two sides proportionally./Verdeel die ander twee sye eweredig.	✓ answer/antwoord	A (1)
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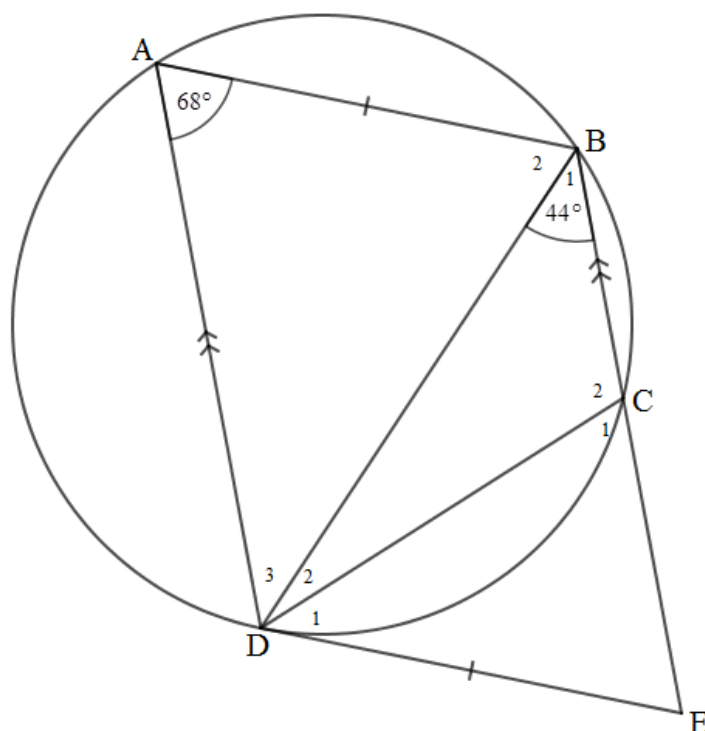
9.2



9.2.1	$\frac{PM}{PQ} = \frac{PN}{PR}$ (Prop. theorem/ewer.st.; $MN \parallel QR$)	✓ ST	A
		✓ RE	A
	$\frac{PM}{35} = \frac{5}{7}$	✓ substitution/verv.	A
	PM = 25 units	✓ length of/lengte van PM	CA

	OR/OF $\frac{PM}{MQ} = \frac{PN}{NR} \quad (\text{Prop. theorem/ewer.st.; } MN \parallel QR)$ $\frac{PM}{35 - PM} = \frac{5}{2}$ $PM = 25 \text{ units}$	OR/OF $\checkmark$ ST A $\checkmark$ RE A $\checkmark$ substitution/verv. A $\checkmark$ length of/lengte van PM A (4)
9.2.2	$\frac{PX}{PQ} = \frac{RY}{RQ} \quad (\text{Prop. theorem/ewer.st.; } XY \parallel PR)$ $\frac{PX}{35} = \frac{1}{4}$ $PX = \frac{35}{4} = 8,75 \text{ units}$ $\therefore XM = PM - PX$ $= 25 - 8,75$ $= 16,25 \text{ units}$ OR/OF $\frac{PX}{XQ} = \frac{PX}{35 - PX} = \frac{RY}{QY} = \frac{1}{3} \quad (\text{Prop. theorem/ewer.st.; } XY \parallel PR)$ $PX = \frac{35}{4} = 8,75 \text{ units}$ $\therefore XM = PM - PX$ $= 25 - 8,75$ $= 16,25 \text{ units}$ OR $\frac{QX}{35} = \frac{3}{4} \quad (\text{prop. theorem/ewer.st.; } XY \parallel PR)$ $QX = 26,25$ $PX = 35 - 26,25 = 8,75$ $XM = 25 - 8,75 = 16,25 \text{ units}$	$\checkmark$ ST A $\checkmark$ RE A $\checkmark$ length of/lengte van PX A $\checkmark$ M A $\checkmark$ Value of/Waarde van XM CA OR/OF $\checkmark$ ST A $\checkmark$ RE A $\checkmark$ length of/lengte van PX A $\checkmark$ M A $\checkmark$ Value of/Waarde van XM CA OR $\checkmark$ ST A $\checkmark$ RE A $\checkmark$ length of/lengte van PX A $\checkmark$ M A $\checkmark$ Value of/Waarde van XM CA (5)

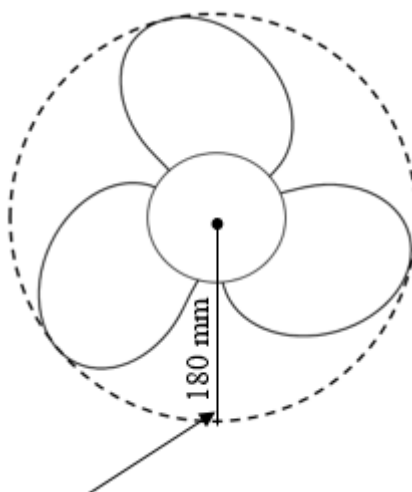
9.3



9.3.1	$\hat{D}_1 = \hat{B}_1 = 44^\circ$ (tan – chord theorem/rklyn-koord st) $\hat{D}_3 = \hat{B}_1 = 44^\circ$ (alt./verw. $\angle$ s; $AD \parallel BC$)	✓ ST ✓ RE ✓ ST ✓ RE	A A A A (4)
9.3.2	$\hat{C}_2 = 180^\circ - 68^\circ = 112^\circ$ (opp. $\angle$ s of cyclic quad./ teenoorst $\angle$ e van kvhk) OR/OF $\hat{C}_2 = 180^\circ - 68^\circ = 112^\circ$ ($\angle$ s on straight line/ reguitlyn)	✓ ST ✓ RE OR/OF ✓ ST ✓ RE	A A OR/OF A A (2)
9.3.3	In $\triangle ABD$ and $\triangle CED$ $\hat{D}_1 = \hat{D}_3$ (proven/bewys) $\hat{A} = \hat{C}_1 = 68^\circ$ (ext. $\angle$ of cyclic quad/ buite $\angle$ van kvhk) $\hat{B}_2 = \hat{E}$ (3rd $\angle$ s of/van Δ s) $\therefore \triangle ABD \parallel \triangle CED$ ($\angle$; $\angle$; $\angle$) or/of equiangular/gelykhoekig	✓ ST ✓ ST/RE ✓ RE	A A A (3)
			[19]

QUESTION/VRAAG 10

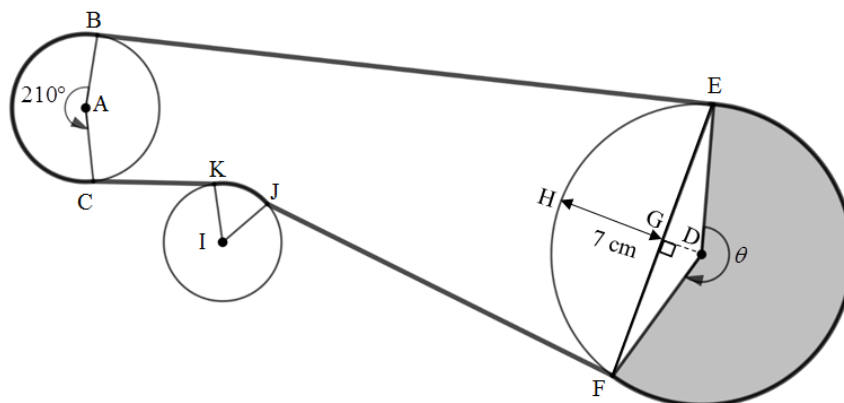
10.1



Radius of the circular path formed
by the rotating propeller blades/
Radius van die sirkelpad gevorm
deur die roterende skroefblaaie

10.1.1	$V = 30 \times \frac{1000}{3600}$ $= \frac{25}{3} \text{ or/of } \square 8,33 \text{ m/s}$	✓ multiplying by/ vermenigvuldig met $\frac{1000}{3600}$ A ✓ 8,33 m/s CA (2)
10.1.2	$V = \omega r$ $\frac{25}{3} = \omega (0,18)$ $\omega = \frac{1\,250}{27} \text{ rad/s or/of } \square 46,30 \text{ rad/s}$ OR/OF $V = \pi D n$ $\frac{25}{3} = \pi (0,36) n$ $n = \frac{625}{27\pi} \text{ rev/s or/of } \square 7,37 \text{ rev/s}$ $\omega = 2\pi n$ $= 2\pi \frac{625}{27\pi} \text{ or/of } \square 2\pi (7,37)$ $= \frac{1250}{27} \text{ rad/s or/of } \square 46,31 \text{ rad/s}$	✓ formula/formule A ✓ converting/herlei r to 0,18 m A ✓ SF CA ✓ angular velocity/ hoeksnelheid CA OR/OF ✓ converting/herlei D to 0,36 m A ✓ rotational frequency/ rotasiesnelheid A ✓ SF CA ✓ angular velocity/ hoeksnelheid CA NPR (4)

10.2

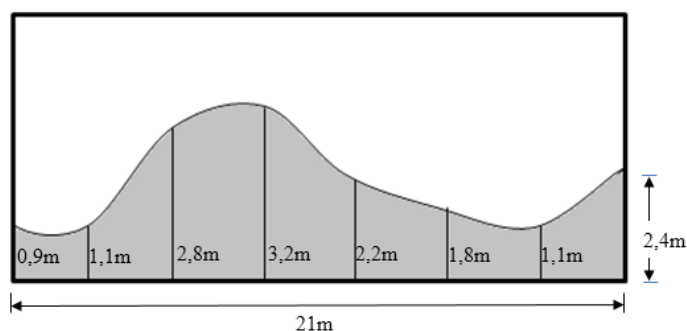


10.2.1	$210^\circ \times \frac{\pi}{180^\circ} = \frac{7}{6}\pi$ or / of $\approx 3,67$ rads	✓ $\frac{7}{6}\pi$ rads	A (1)
10.2.2	$s = r \theta$ $= (5) \left(\frac{7}{6}\pi \right)$ $= \frac{35}{6}\pi \approx 18,33$ cm	✓ formula/formule ✓ SF ✓ arc length BC/booglengte	A CA CA (3)
10.2.3	$A = \frac{r^2 \theta}{2}$ $54\pi = \frac{(9)^2 \theta}{2}$ $\theta = \frac{2(54\pi)}{(9)^2}$ $\theta = \frac{4}{3}\pi$ rad or/of = 240° or/of $\approx 1,33\pi$ rad or/of $\approx 4,19$ rad OR/OF $A = \frac{rs}{2}$ $54\pi = \frac{9s}{2}$ $s = 12\pi$ or/of $\approx 37,70$ cm $\theta = \frac{s}{r}$ $= \frac{12\pi}{9}$ or/of $\approx \frac{37,70}{9}$ $= \frac{4}{3}\pi$ rad or/of $\approx 4,19$ rad	✓ formula/formule ✓ SF ✓ θ OR/OF ✓ formula/formule ✓ SF ✓ θ	A A CA A A CA (3)

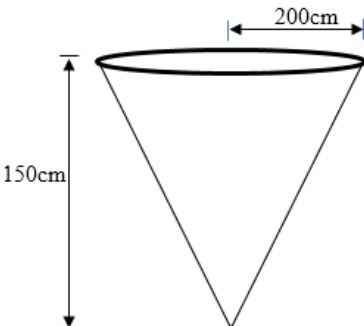
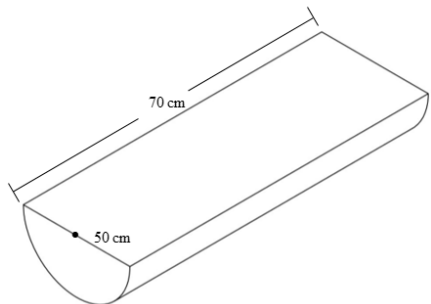
10.2.4	$4h^2 - 4dh + x^2 = 0$ $4(7)^2 - 4(18)(7) + x^2 = 0$ $x = \sqrt{4(18)(7) - 4(7)^2}$ $\therefore x = EF = 2\sqrt{77} \approx 17,55 \text{ cm}$ OR/OF Half-chord method/Halfkoord metode: $GD = DH - GH = 9 - 7 = 2 \text{ cm}$ $GE = \sqrt{9^2 - 2^2} = \sqrt{77}$ (Pythagoras Thm.) $EF = 2GE = 2\sqrt{77} \approx 17,55 \text{ cm}$ OR/OF $\hat{FDE} = 154,32^\circ$ $FE = \sqrt{FD^2 + DE^2 - 2FD \cdot DE \cos \hat{FDE}}$ $= \sqrt{(9)^2 + (9)^2 - 2(9)(9) \cos 154,32^\circ}$ $\square \sqrt{307,9989914}$ $\square 17,55 \text{ m}$	✓ formula/formule A ✓ SF A ✓ length of chord/lengte van koord CA OR/OF ✓ length of/lengte van GD A ✓ length of/lengte van GE A ✓ length of/lengte van EF CA OR/OF ✓ cos rule/reël A ✓ SF A ✓ length of/lengte van EF CA (3)
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10.2.5	Length of major arc/<i>Lengte van grootboog EF</i>: $s = r \theta$ $= (9) \left(\frac{4}{3} \pi \right)$ $= 12\pi \approx 37,70 \text{ cm}$ OR/ OF $A = \frac{rs}{2}$ $54\pi = \frac{9s}{2}$ $s = 12\pi \approx 37,70 \text{ cm}$ Length of rubber belt not in contact with pulleys/ <i>Lengte van rubberband wat nie met katrolle kontak maak nie</i>: ☐ $140 - (4,19 + 18,33 + 37,70)$ ☐ $140 - 60,21$ ☐ $79,78 \text{ cm}$	✓ F A ✓ SF CA ✓ length of arc/<i>lengte van grootboog EF</i> CA ✓ M A ✓ length/<i>lengte</i> CA (5)
		[21]

QUESTION/ VRAAG 11



11.1.1	$a = \frac{21}{7} = 3\text{m}$ $A_T = a \left(\frac{o_1 + o_n}{2} + o_2 + o_3 + \dots + o_{n-1} \right)$ $= 3 \left(\frac{0,9 + 2,4}{2} + 1,1 + 2,8 + 3,2 + 2,2 + 1,8 + 1,1 \right) \text{m}^2$ $= 3 (1,65 + 1,1 + 2,8 + 3,2 + 2,2 + 1,8 + 1,1) \text{m}^2$ $= 41,55 \text{ m}^2$ OR/OF $a = \frac{21}{7} = 3\text{m}$ $A_T = a (m_1 + m_2 + m_3 + \dots + m_n)$ $= 3 \left(\frac{0,9 + 1,1}{2} + \frac{1,1 + 2,8}{2} + \frac{2,8 + 3,2}{2} + \frac{3,2 + 2,2}{2} + \frac{2,2 + 1,8}{2} + \frac{1,8 + 1,1}{2} + \frac{1,1 + 2,4}{2} \right) \text{m}^2$ $= 3 (1 + 1,95 + 3 + 3 + 2,7 + 2 + 1,45 + 1,75) \text{m}^2$ $= 41,55 \text{ m}^2$	✓ value of/waarde van a A ✓ formula/formule A ✓ SF CA ✓ value of/ waarde van A_T CA OR/OF ✓ value of/waarde van a A ✓ F A ✓ SF CA ✓ value of/waarde van A_T CA (4)
11.1.2	$A_T = 41,55 \times \frac{180}{100}$ $= 74,79 \text{ m}^2$ OR/OF $21 \times \frac{180}{100} = 37,8\text{m}$ $a = 5,4\text{m}$ $A_T = 5,4 \left(\frac{0,9 + 2,4}{2} + 1,1 + 2,8 + 3,2 + 2,2 + 1,8 + 1,1 \right)$ $= 74,79 \text{ m}^2$	✓ $\frac{180}{100}$ A ✓ value of area CA OR/OF ✓ value of/waarde van a A ✓ value of area/waarde van oppervlakte CA (2)

11.2	 	
11.2.1	Exterior surface area = $\frac{1}{2}(2\pi r^2 + 2\pi rh)$ $= \frac{1}{2}(2\pi (50)^2 + 2\pi (50)(70))$ $= 6000\pi \text{ or / of } \square 18849,56 \text{ cm}^2$ OR/OF TSA = $\pi r^2 + \pi rh$ $= \pi(50)^2 + \pi(50)(70)$ $\square 18\,849,56 \text{ cm}^2$ OR/OF A = $2\pi r^2 + 2\pi rh$ $= 2\pi(50)^2 + 2\pi(50)(70)$ $\square 37\,699,12 \text{ cm}^2$ $\therefore \text{TSA } \square \frac{1}{2}(37\,699,12) \square 18\,849,56$	✓F A ✓SF A ✓ value of/waarde van TSA CA OR/OF ✓F A ✓SF A ✓ value of/waarde van TSA CA OR/OF ✓F A ✓SF A ✓ value of/waarde van TSA CA NPR (3)
11.2.2	Volume of cone = $\frac{1}{3}(\pi r^2 h)$ $= \frac{1}{3}(\pi (200)^2 (150))$ $= 2000000\pi \text{ cm}^3 \text{ or / of } \square 6\,283\,185,307 \text{ cm}^3$ Volume of half cylinder = $\frac{1}{2}(\pi r^2 h)$ $= \frac{1}{2}(\pi (50)^2 (70))$ $= 87500\pi \text{ cm}^3 \text{ or / of } \square 274889,3572 \text{ cm}^3$ number of times = $\frac{2000000\pi \text{ cm}^3}{87500\pi \text{ cm}^3} = 22,86$ He can fully fill the half-cylindrical tank 22 times./ Hy kan die halfsilindriese tenk 22 keer heeltemal vol maak.	✓SF A ✓ value of/waarde van V CA ✓SF A ✓ value of/waarde van V CA ✓ 22 CA PR (5)
		[14]

